TEXTBOOK EXERCISES

I. Choose the correct answer.

1. All the factors of biosphere which affect the ability of organisms to survive and reproduce are called as _____.
   a. biological factors b. abiotic factors
   c. biotic factors d. physical factors
2. The ice sheets from the north and south poles and the icecaps on the mountains, get converted into water vapour through the process of_____.
   a. evaporation b. condensation
   c. sublimation d. infiltration
3. The atmospheric carbon dioxide enters into the plants through the process of _____.
   a. photosynthesis b. assimilation
   c. respiration d. decomposition
4. Increased amount of _____ in the atmosphere, results in greenhouse effect and global warming
   a. carbon monoxide
   b. sulphur dioxide
   c. nitrogen dioxide
   d. carbon dioxide

II. Match the following.

Microorganism	Role Played
Nitrosomonas	Nitrogen fixation
Azotobacter	Ammonification
Pseudomonas species	Nitrification
Putrefying bacteria	Denitrification

III. State whether true or false. If false, correct the statement.

1. Nitrogen is a greenhouse gas.
2. Poorly developed root is an adaptation of mesophytes.
3. Bats are the only mammals that can fly.
4. Earthworms use the remarkable high frequency system called echoes.
5. Aestivation is an adaptation to overcome cold condition.

IV. Give reason for the following.

1. Roots grow very deep and reach the layers where water is available. Which type of plants develops the above adaptation? Why?
2. Why streamlined bodies and presence of setae is considered as adaptations of earthworm?
3. Echolocation is the adaptation of bats. Is this correct?

V. Answer briefly.

1. What are the two factors of biosphere?
2. How do human activities affect nitrogen cycle?
3. What is adaptation?
4. What are the challenges faced by hydrophytes in their habitat?
5. Why is it important to conserve water?
6. List some of the ways in which you could save water in your home and school.
7. What are the uses of recycled water?
8. What is IUCN? What is the vision of IUCN?

VI. Answer in detail.

1. Describe the processes involved in the water cycle.
2. Explain carbon cycle with the help of a flow chart.
3. List out the adaptations of xerophytes.
4. How does a bat adapt itself to its habitat?
5. What is water recycling? Explain the conventional wastewater recycling treatment methods.

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